# Assignment

accredian
credentials that matter

# Instructor Effectiveness Modeling (EdTech Context)

**Role:** Data Science / AI Content Specialist Intern

**Submission Format:** Google Colab or Jupyter Notebook

**Allowed Libraries:** Python, pandas, numpy, scikit-learn, matplotlib / seaborn

**Not Allowed:** LLMs, external datasets, AutoML tools

---

## 1. Problem Context

You are working with an **EdTech platform** that runs the *same course across multiple batches* taught by different instructors.

Each instructor may teach:

- Multiple batches
- The same course across different batches
- Different courses across time

The company wants to better understand **instructor effectiveness** using available data from learner outcomes, engagement, and feedback.

Your task is to **analyze the data, define instructor effectiveness, and build an ML model** to predict instructor effectiveness tiers.

There is **no single correct answer**.
We are more interested in **how you think** than the final model accuracy.

---

## 2. Dataset Description

You are provided with a CSV file where:

- **Each row represents one course batch**
- Multiple rows may belong to the same instructor

### 2.1 Identifier Columns

| Column | Description |
|---|---|
| batch_id | Unique ID for a course batch |
| instructor_id | Unique instructor identifier |
| course_id | Course identifier |

---

### 2.2 Learner Outcome Metrics

| Column | Description |
|---|---|
| completion_rate | Fraction of learners who completed the course (0–1) |
| dropout_rate | Fraction of learners who dropped out (0–1) |
| avg_score_improvement | Average improvement from pre- to post-assessment |

| | |
|---|---|
| avg_quiz_score | Average quiz score for the batch |

---

### 2.3 Engagement Metrics

| Column | Description |
|---|---|
| avg_watch_time | Normalized average video watch time (0–1) |
| assignment_submission_rate | Fraction of learners submitting assignments |
| forum_activity_rate | Fraction of learners active on discussion forums |

---

### 2.4 Feedback Metrics

| Column | Description |
|---|---|
| avg_feedback_score | Average learner feedback rating (1–5) |
| feedback_response_rate | Fraction of learners who submitted feedback |

---

## 3. Your Objective

Your overall goal is to **estimate instructor effectiveness** using machine learning.

You will need to:

1. Define what “instructor effectiveness” means
2. Aggregate batch-level data to instructor-level
3. Build an ML model to predict instructor effectiveness tiers
4. Analyze and interpret your results

---

## 4. Tasks to Complete

**Step 1: Exploratory Data Analysis (EDA)**

- Explore distributions and relationships between variables
- Identify correlations, missing values, or anomalies
- Highlight any early observations

📌 *Use plots and short explanations.*

---

**Step 2: Define Instructor Effectiveness**

You must define an **Instructor Effectiveness Score** using the available data.

Then convert this score into **effectiveness tiers**, for example:

- Low
- Medium
- High

⚠️ There is no predefined formula.
You must **justify your definition** and assumptions.

---

**Step 3: Aggregate Batch Data to Instructor Level**

Since instructors teach multiple batches:

- Aggregate batch-level features into instructor-level features
- Decide how to handle instructors with:
    - Few batches
    - Many batches

Explain:

- Which aggregation functions you used
- Why they make sense

---

**Step 4: Build a Machine Learning Model**

- Train a model to predict instructor effectiveness tier
- You may choose any **classical ML algorithm**
- Feature selection and scaling are optional but encouraged

📌 *Model choice matters less than your reasoning.*

---

**Step 5: Evaluate the Model**

- Choose appropriate evaluation metrics

- Discuss trade-offs (e.g., precision vs recall)
- Comment on class imbalance if present

---

**Step 6: Interpret the Results**

- Identify key features influencing instructor effectiveness
- Explain results in simple, non-technical terms
- Suggest how insights could be used in an EdTech product

---

## 5. Mandatory Analysis Questions

*(Answer in markdown cells in the notebook)*

1. Which features most influenced instructor effectiveness, and why?
2. Which variables could be misleading or confounded?
3. How could this model fail in real-world usage?
4. What additional data would you want to improve this analysis?
5. Should this model be used for instructor performance evaluation? Why or why not?

---

## 6. Constraints & Guidelines

- ❌ Do not use LLMs or external APIs
- ❌ Do not use external datasets
- ✅ Focus on clarity and reasoning
- ✅ Keep the notebook reproducible
- ✅ Comment your code and decisions

Accuracy is **not the primary evaluation metric**.

---

## 7. What We Are Evaluating

We will assess your submission on:

- Understanding of the problem
- Quality of feature engineering & aggregation
- ML fundamentals and evaluation logic
- Interpretation and communication
- Awareness of limitations and ethical considerations

---

## 8. Submission Instructions

- Submit a **single notebook file**
- Ensure all cells run without errors
- Include explanations in markdown cells
- Clearly label sections

---

## 9. Final Note to Candidates

This task reflects real problems faced by EdTech companies.

Strong submissions:

- Clearly explain assumptions
- Discuss trade-offs
- Acknowledge limitations
- Demonstrate structured thinking

There is **no penalty for a simple model** if the reasoning is strong.

---

**Best of luck — we're excited to see how you think.**